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United States Patent	12413042
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Yamada; Kohei

Semiconductor laser element, light emitting device, and method for manufacturing these

Abstract

A semiconductor laser element includes a semiconductor-containing part, an electrode and at least one metal film. The semiconductor-containing part has first and second main surfaces, a light emitting end surface, a light reflecting end surface, and an optical waveguide. A distance between the first main surface and the optical waveguide is greater than a distance between the second main surface and the optical waveguide. The electrode is provided on the first main surface. The metal film is provided on the first main surface at a position spaced apart from the electrode. The metal film is in contact with a first side of an outer edge of the first main surface on a side of the light emitting end surface. The metal film is arranged at a position that does not overlap the optical waveguide in a plan view seen along a normal direction of the first main surface.

Inventors:	Yamada; Kohei (Komatsushima, JP)
Applicant:	NICHIA CORPORATION (Anan, JP)
Family ID:	83286072
Assignee:	NICHIA CORPORATION (Anan, JP)
Appl. No.:	17/699068
Filed:	March 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220302670 A1	Sep. 22, 2022

Foreign Application Priority Data

JP	2021-045643	Mar. 19, 2021
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Publication Classification

Int. Cl.: **H01S5/0238** (20210101); **H01S5/02** (20060101); **H01S5/026** (20060101); **H01S5/028** (20060101); **H01S5/042** (20060101)

U.S. Cl.:

CPC **H01S5/026** (20130101); **H01S5/0202** (20130101); **H01S5/028** (20130101); **H01S5/04256** (20190801);

Field of Classification Search

CPC: H01S (5/026); H01S (5/0202); H01S (5/028); H01S (5/04256); H01S (5/0014); H01S (5/0234); H01S (5/0282); H01S (5/32341); H01S (5/0238); H01S (5/02255); H01S (5/22); H01S (2301/176)

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Primary Examiner: Niu; Xinning (Tom)

Attorney, Agent or Firm: Global IP Counselors, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Japanese Patent Application No. 2021-045643, filed on Mar. 19, 2021, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) The present disclosure relates to a semiconductor laser element, a light emitting device, and a method for manufacturing these.

(3) In Japanese Laid-Open Patent Application Publication No. 2004-095859, noted is a method for manufacturing a semiconductor laser that obtains individual semiconductor lasers by forming a plurality of electrode patterns that are smaller than the chip size that fits within the chip size of a resonator length, and dividing between the electrode patterns. In Japanese Laid-Open Patent Application Publication No. 2004-095859, with this method, since it is difficult to produce a laser chip of different resonator lengths from the same semiconductor wafer, as a solution, noted is a method for manufacturing a semiconductor laser that forms the electrode patterns continuously in the resonator length direction. With this method, individual semiconductor lasers are obtained by cleaving at a position that crosses the electrode patterns.

SUMMARY

(4) With the laser elements obtained by dividing between electrode patterns, since there is no electrode pattern on the end surface of the laser elements, there are cases when it is not possible to determine the position of the end surface of the laser element using an image recognition device, etc. In this case, for example, the electrode pattern is recognized using an image recognition device, etc., and the end surface is estimated to be at a position at a prescribed distance from that electrode pattern. It is possible to perform various types of pass/fail determination of the position at which the laser element is mounted based on such an estimate, for example, but since the actual position of the end surface does not necessarily match the estimated position, there is a possibility that an item that is originally a good item will be erroneously determined as a defective item.

(5) Also, when cleaving is performed at a position that crosses the electrode patterns as with the manufacturing method noted in Japanese Laid-Open Patent Application Publication No. 2004-095859, the electrode pattern is formed up to the end part obtained by cleaving, but on the other hand, there is a risk of causing a short circuit, etc., due to the electrode pattern stretching due to cleaving.

(6) The present disclosure includes the following embodiments according to one aspect of the present invention.

(7) A semiconductor laser element according to one aspect of the present invention includes a semiconductor-containing part, an electrode and at least one metal film. The semiconductor-containing part has a first main surface, a second main surface, a light emitting end surface, a light reflecting end surface, and an optical waveguide. A distance between the first main surface and the optical waveguide is greater than a distance between the second main surface and the optical waveguide. The electrode is provided on the first main surface. The at least one metal film is provided on the first main surface at a position spaced apart from the electrode. An outer edge of the first main surface has a first side on a side of the light emitting end surface. The at least one metal film is in contact with the first side of the outer edge of the first main surface. The at least one metal film is arranged at a position that does not overlap the optical waveguide in a plan view seen along a normal direction of the first main surface.

(8) A light emitting device according to one aspect of the present invention includes the semiconductor laser element described above, and a base on which the semiconductor laser element is fixed. The semiconductor laser element is fixed to the base in an arrangement in which the second main surface of the semiconductor-containing part faces the base.

(9) A method for manufacturing a semiconductor laser element according to one aspect of the present invention includes: preparing a semiconductor-containing part having a first main surface, a second main surface, and an optical waveguide, a distance between the first main surface and the optical waveguide being greater than a distance between the second main surface and the optical waveguide; forming a plurality of electrodes and at least one metal film on the first main surface of the semiconductor-containing part so that the electrodes are aligned in a first direction, and the at

least one metal film is arranged at a position that is between the electrodes and spaced apart from the electrodes, and at a position that does not overlap with the optical waveguide in a plan view when seen along a normal direction of the first main surface; and cleaving the semiconductor-containing part at a position between the electrodes along a second direction intersecting the first direction thereby dividing the at least one metal film.

(10) A method for manufacturing a light emitting device according to one aspect of the present invention includes preparing a semiconductor laser element using the method described above, and fixing the semiconductor laser element to a base in an arrangement in which the second main surface of the semiconductor-containing part faces the base.

(11) With the semiconductor laser element, the light emitting device, and the method for manufacturing these described above, it is possible to improve the pass/fail determination precision.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a schematic plan view showing a semiconductor laser element of an embodiment according to the present invention.

(2) FIG. 2A is a cross section view of line IIA-IIA of FIG. 1.

(3) FIG. 2B is a cross section view of line IIB-IIB of FIG. 1.

(4) FIG. 3 is a schematic plan view showing another example of a semiconductor laser element of an embodiment according to the present invention.

(5) FIG. 4A is a schematic plan view showing a light emitting device of an embodiment according to the present invention.

(6) FIG. 4B is a schematic cross section view showing another example of the light emitting device of an embodiment according to the present invention.

(7) FIG. 5 is a flow chart showing a method for manufacturing a semiconductor laser element of an embodiment according to the present invention.

(8) FIG. 6 is a partial schematic plan view showing the method for manufacturing a semiconductor laser element of an embodiment according to the present invention.

(9) FIG. 7 is a cross section view of line VII-VII of FIG. 6.

(10) FIG. 8 is a partial schematic plan view showing the method for manufacturing a semiconductor laser element of the embodiment.

(11) FIG. 9 is a cross section view of line IX-IX of FIG. 8.

(12) FIG. 10 is a partial schematic plan view showing the method for manufacturing a semiconductor laser element of the embodiment.

(13) FIG. 11 is a flow chart showing the method for manufacturing a light emitting device of an embodiment according to the present invention.

DETAILED DESCRIPTION

(14) Hereafter, embodiments of the present invention are explained while referring to the drawings. In each drawing, the same code number is given to the same element. Also, in each cross section view, only the state of the cross section surface is shown, and illustrations of members that do not exist in that cross section surface are omitted.

(15) FIG. 1 is a schematic plan view showing a semiconductor laser element of an embodiment. FIG. 2A is a cross section view of line IIA-IIA of FIG. 1. FIG. 2B is a cross section view of line IIB-IIB of FIG. 1. As shown in FIG. 1 to FIG. 2B, the semiconductor laser element **100** has a semiconductor-containing part **10**, an electrode **20**, and one or more metal films **30**.

(16) (Semiconductor-Containing Part **10**)

(17) The semiconductor-containing part **10** has a first main surface **11**, a second main surface **12**, a

light emitting end surface **13**, a light reflecting end surface **14**, and an optical waveguide **15**. The outer edge of the first main surface **11** has a first side **11a** on the side of the light emitting end surface **13**. The outer edge of the first main surface **11** has a second side **11b** on the side of the light reflecting end surface **14**. The outer edge of the first main surface **11** is a quadrangle, for example, in the plan view seen along the normal direction of the first main surface **11**. The outer edge of the second main surface **12** is a quadrangle, for example, in the plan view seen along the normal direction of the second main surface **12**. For example, the first main surface **11** is a flat plane, and a ridge **17d** is provided on the second main surface **12**. If a metal film **30** is provided on a part with a level difference, contrast occurs due to the level difference, so there is a possibility that the level difference will be erroneously detected as the outer shape of the metal film **30** by an image recognition device, etc. For this reason, it is preferable that the metal film **30** be provided on a flat part. If the first main surface **11** is a flat plane, there is no level difference regardless of where the metal film **30** is formed, so it is possible to improve the degree of freedom for the formation position of the metal film **30**.

(18) The distance between the first main surface **11** and the optical waveguide **15** is greater than the distance between the second main surface **12** and the optical waveguide **15**. As a result, when the semiconductor laser element **100** is fixed to a base in an arrangement in which the second main surface **12** of the semiconductor laser element **100** faces the base as described later, it is possible to detect the positional relationship of the light emitting end surface **13** and the base using the metal film **30**. Therefore, it is possible to improve the precision of the pass/fail determination of their positional relationship. The size relationship between the distance from the first main surface **11** to the optical waveguide **15** and the distance from the second main surface **12** to the optical waveguide **15** may be determined based on the size relationship between the distance from the first main surface **11** to an active layer **17b** and the distance from the second main surface **12** to the active layer **17b**. Specifically, if the distance between the first main surface **11** and the active layer **17b** is greater than the distance between the second main surface **12** and the active layer **17b**, it can be said that the distance between the first main surface **11** and the optical waveguide **15** is greater than the distance between the second surface **12** and the optical waveguide **15**.

(19) The optical waveguide **15** is arranged at a position connecting the light emitting end surface **13** and the light reflecting end surface **14**. The optical waveguide **15** is a shape that is long in one direction, for example. The optical waveguide **15** is a shape that is long in the direction in which the light emitting end surface **13** and the light reflecting end surface **14** intersect, for example. The long direction of the optical waveguide **15** can intersect the light emitting end surface **13** and the light reflecting end surface **14** at an angle of 80 to 100°. The short direction of the optical waveguide **15** is the direction that intersects perpendicular to the long direction in the plan view.

(20) The semiconductor-containing part **10** can have a substrate **16** and a semiconductor layer **17**. The substrate **16** has a first substrate main surface **16a** and a second substrate main surface **16b** on the side opposite to the first substrate main surface **16a**. The semiconductor layer **17** is provided on the second substrate main surface **16b** of the substrate **16**.

(21) The substrate **16** is a semiconductor substrate, for example. A nitride semiconductor substrate such as a GaN substrate can be used as the substrate **16**. The easy cleaving surface of a nitride semiconductor having a wurtzite structure of GaN, etc., is the m surface (specifically, {10-10} surface). For example, it is possible to form the semiconductor layer **17** on +c surface (specifically, the (0001) surface) of the substrate **16**, and to use m surfaces of the substrate **16** and the semiconductor layer **17** as the light emitting end surface **13** and the light reflecting end surface **14**. The +c surface is not limited to being a surface that strictly matches the (0001) surface, and also includes surfaces having an off angle within a range of ± 1 degree. The m surface is not limited to being a surface that strictly matches the {10-10} surface, and also includes surfaces for which the skew from the {10-10} surface is less than 1 degree. It is also possible to not provide the substrate **16**, and to have the semiconductor-containing part **10** consist only of the semiconductor layer **17**.

(22) The semiconductor layer **17** has a first electrically conductive semiconductor layer **17a**, an active layer **17b**, and a second electrically conductive semiconductor layer **17c**. The semiconductor constituting these can be a nitride semiconductor, and can be a group **111** nitride semiconductor, for example. Examples of a group III nitride semiconductor include GaN, InGaN, and AlGaN.

(23) The first electrically conductive semiconductor layer **17a** is an n-type semiconductor layer, for example. The second electrically conductive semiconductor layer **17c** is a p-type semiconductor layer, for example. The substrate **16**, the first electrically conductive semiconductor layer **17a**, the active layer **17b**, and the second electrically conductive semiconductor layer **17c** may each be in direct contact, or may have a separate semiconductor layer arranged therebetween. For example, it is possible to have an undoped layer arranged between the second electrically conductive semiconductor layer **17c** and the active layer **17b**. The active layer **17b** can have a multiple quantum well structure or a single quantum well structure. As the plurality of layers formed on the substrate **16**, examples include in order from the substrate **16** side an n-side clad layer, an n-side light guide layer, the active layer **17b**, a p-side electron confinement layer, a p-side light guide layer, a p-side clad layer, and a p-side contact layer. For example, the n-side clad layer is the first electrically conductive semiconductor layer **17a**, and the p-side contact layer is the second electrically conductive semiconductor layer **17c**.

(24) The semiconductor layer **17** has the ridge **17d** provided on the surface of the side opposite to the substrate **16**. In the plan view, the ridge **17d** is arranged at a position that connects the light emitting end surface **13** and the light reflecting end surface **14**. The ridge **17d** has a long shape in one direction, for example. The ridge **17d** has a long shape in the direction intersecting the light emitting end surface **13** and the light reflecting end surface **14**, for example. The long direction of the ridge **17d** can intersect the light emitting end surface **13** and the light reflecting end surface **14** at an angle of 80 to 100°. The ridge **17d** can be a stripe shape, for example. The optical waveguide can be constituted by the ridge **17d**. For example, in the plan view seen along the normal direction of the first main surface **11**, the part that matches the ridge **17d** can be regarded as the optical waveguide **15**. The ridge **17d** is formed on a portion of the top surface of the second electrically conductive semiconductor layer **17c**, for example.

(25) (Electrode **20**)

(26) The electrode **20** is provided on the first main surface **11**. In FIG. **1** to FIG. **2B**, the first main surface **11** is configured by the first substrate main surface **16a** of the substrate **16**. Specifically, the electrode **20** is provided on the first substrate main surface **16a**. The electrode **20** is an n electrode, for example. The electrode **20** has one or more layers of a layer of one or more metals including Ni, Rh, Cr, Au, W, Pt, Ti, Al or an alloy, for example. The electrode **20** has an Au layer as the outermost layer, for example. The electrode **20** is not in contact with the first side **11a** of the first main surface **11**. In the plan view, the distance between the electrode **20** and the first side **11a** can be 15 μm or greater, or can be 45 μm or less. The electrode **20** is not in contact with the second side **11b** of the first main surface **11**. In the plan view, the distance between the electrode **20** and the second side **11b** may also be 15 μm or greater. The distance between the electrode **20** and the second side **11b** can be 45 μm or less. The electrode **20** and the second side **11b** may also be in contact. In FIG. **1**, the electrode **20** is not arranged between the plurality of metal films **30**, but it is also possible to arrange a portion of the electrode between the plurality of metal films **30**. At least a portion of the electrode **20** can be arranged at a position that overlaps the optical waveguide **15** in the plan view seen along the normal direction of the first main surface **11**.

(27) (Metal Film **30**)

(28) The metal film **30** is provided on the first main surface **11**. The metal film **30** is provided at a position spaced apart from the electrode **20**. The metal film **30** is in contact with the first side **11a** of the first main surface **11**. The metal film **30** fulfills at least one of (a) and (b) below. It may also satisfy both. (a) A length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** is smaller than a length $L_{\text{sub.20}}$ in the direction parallel to the first side **11a** of the electrode **20**. (b) The metal film

30 is arranged at a position that does not overlap with the optical waveguide **15** in the plan view when seen along the normal direction of the first main surface **11**.

(29) By providing the metal film **30**, it is possible to improve the pass/fail determination precision. Specifically, because the metal film **30** is in contact with the first side **11a** of the first main surface **11**, it is possible to determine the position of the light emitting end surface **13** using the metal film **30**. This makes it possible to do pass/fail determination based on the actual position of the light emitting end surface **13**, so it is possible to improve the pass/fail determination precision. Also, after fixing the semiconductor laser element **100** to the base described later, the position of the light emitting end surface **13** is specified by detecting the outline of the metal film **30** using an image recognition device, etc., and it is possible to perform pass/fail determination of the fixing position of the semiconductor laser element **100** from the positional relationship of the end part of the base and the light emitting end surface **13**.

(30) Also, by providing the metal film **30**, it is possible to reduce the probability that the end surface protective film **62** shown in FIG. 3 described later will reach the electrode **20**. This is because there is the metal film **30** between from the light emitting end surface **13** to the electrode **20**, and using the metal film **30**, it is possible to reduce the wraparound of the end surface protective film **62** provided on the light emitting end surface **13**. In this case, it is preferable the first main surface **11** be a flat plane, and that the thickness of the metal film **30** and the thickness of the electrode **20** be equal, or that the thickness of the metal film **30** be larger. This makes it possible to further reduce the possibility of the end surface protective film **62** provided on the light emitting end surface **13** from reaching the electrode **20**. In particular, when forming the end surface protective film **62**, when arranging a spacer on the side of the first main surface **11**, if the thickness of the metal film **30** and the thickness of the electrode **20** are in that kind of relationship, it is possible to further reduce wraparound of the end surface protective film **62** using the metal film **30** and the spacer.

(31) The metal film **30** is not connected with the electrode **20**. For this reason, compared to when the electrode **20** is formed in contact with the first side **11a**, there is a lower possibility of a short circuit occurring when forming the light emitting end surface **13**. The distance between the electrode **20** and the metal film **30** can be 5 μm or greater. This makes it possible to more reliably separate the electrode **20** and the metal film **30**. The distance between the electrode **20** and the metal film **30** can be 20 μm or less.

(32) The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** is smaller than the length $L_{\text{sub.20}}$ in the direction parallel to the first side **11a** of the electrode **20**. This makes it possible to further reduce the possibility of a short circuit occurring. It is also possible to reduce the probability that the light emitting end surface **13** will be covered by the metal film **30**. The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** indicates the length of the portion in contact with the first side **11a** of the metal film **30**. The length $L_{\text{sub.20}}$ in the direction parallel to the first side **11a** of the electrode **20** indicates the item that is largest among the lengths in the direction parallel to the first side **11a** of the electrode **20**. The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** may also be smaller than the length in the direction parallel to the first side **11a** of the end part of the light emitting end surface **13** side of the electrode **20**.

(33) The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** can be 90% or less than the length $L_{\text{sub.20}}$ in the direction parallel to the first side **11a** of the electrode **20**, is preferably 70% or less, and is more preferably 30% or less. This makes it possible to reduce the possibility of a short circuit occurring. It is also possible to reduce the probability of the light emitting end surface **13** being covered by the metal film **30**. The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** can be 10% or greater than the length $L_{\text{sub.20}}$ in the direction parallel to the first side **11a** of the electrode **20**.

(34) The length $L_{\text{sub.30}}$ along the first side **11a** of the metal film **30** can be 10 μm or greater, and is preferably 20 μm or greater. This makes it possible to improve the precision of image recognition

of the metal film **30**. The length L.sub.30 along the first side **11a** of the metal film **30** can be 100 μm or less. This makes it possible to reduce the possibility of a short circuit occurring.

(35) The longer the length L.sub.30 along the first side **11a** of the metal film **30**, the more it is possible to improve the precision of the position specification of the light emitting end surface **13** by doing image recognition of the metal film **30**. On the other hand, to reduce the possibility of a short circuit or to reduce the probability of the light emitting end surface **13** being covered by the metal film **30**, the shorter the length L.sub.30 along the first side **11a** of the metal film **30** the better. To achieve both of these, it is preferable to provide a plurality of metal films **30**. In FIG. 1, two metal films **30** are provided.

(36) The total of the lengths L.sub.30 along the first side **11a** of the plurality of metal films **30** is preferably 30% or greater than a length L_{1a} of the first side **11a**. This makes it possible to improve the precision of position specification of the light emitting end surface **13** using the metal films **30**. The total of the lengths L.sub.30 along the first side **11a** of the plurality of metal films **30** can be less than the length L.sub.11a of the first side **11a**. A distance D.sub.30 from one end to the other end of the direction along the first side **11a** of the plurality of metal films **30** is preferably half or greater than the length L_{1a} of the first side **11a**, and more preferably $\frac{2}{3}$ or greater. This makes it possible to improve the precision of position specification of the light emitting end surface **13** using the metal film **30**. The distance D.sub.30 indicates the distance from the farthest apart ends of the portions of the plurality of metal films **30** in contact with the first side **11a**. The distance D.sub.30 can be less than the length L.sub.11a of the first side **11a**. The distance D.sub.30 may be less than the length L.sub.20 in the direction parallel to the first side **11a** of the electrode **20**, and may be the length L.sub.20 or greater than the direction parallel to the first side **11a** of the electrode **20**. The shortest distance between adjacent metal films **30** among the plurality of metal films **30** is greater than the shortest distance between the metal film **30** and the electrode **20**, for example.

(37) As shown in FIG. 1, the metal films **30** are arranged at positions that do not overlap with the optical waveguide **15** in the plan view. This makes it possible to reduce the probability that the light emitting end surface **13** will be covered by the metal films **30**. In the plan view, the plurality of metal films **30** can include one set of metal films **30** arranged at a position sandwiching the optical waveguide **15**. In the plan view, one or more of the plurality of metal films **30** can be arranged in one region of two regions partitioned by the optical waveguide **15** in the first main surface **11**, and another one or more of the plurality of metal films **30** can be arranged in the other region. This makes it possible to have the distance D.sub.30 from one end to the other end in the direction along the first side **11a** of the plurality of metal films **30** be longer than when arranging the plurality of metal film **30** only in one region, and possible to improve the precision of the position specification of the light emitting end surface **13** using the metal films **30**. In FIG. 1, one metal film **30** is arranged in one region of the two regions partitioned by the optical waveguide **15** in the first main surface **11**, and the other metal film **30** is arranged in the other region.

(38) The plurality of metal films **30** may be arranged with bilateral symmetry using something as a reference, or may not be arranged with bilateral symmetry. With the semiconductor laser element **100** shown in FIG. 1, the plurality of metal films **30** are arranged with bilateral symmetry sandwiching the optical waveguide **15** in the plan view seen along the normal direction of the first main surface **11**. In FIG. 1, in the plan view, the optical waveguide **15** is arranged overlapping the connecting line of the center point of the first side **11a** and the center point of the second side **11b**. In this case, by arranging the plurality of metal films **30** with bilateral symmetry sandwiching the optical waveguide **15** in this way, it is possible to arrange the metal films **30** efficiently. When providing the optical waveguide **15** at a position that does not overlap with the connecting line of the center point of the first side **11a** and the center point of the second side **11b**, the length L.sub.30 of the metal film **30** provided at the side having the smaller area of the left side and right side of the optical waveguide **15** is preferably smaller than the length L.sub.30 of the metal film **30** provided at the other.

(39) The outer edge of the metal film **30** is a quadrangle, for example, in the plan view. The length along the long direction of the optical waveguide **15** of the metal film **30** can be 10 μm or greater, and more preferably is 15 μm or greater. This makes it possible to improve the precision of image recognition of the metal film **30**. The length along the long direction of the optical waveguide **15** of the metal film **30** can be less than 40 μm , or can be 35 μm or less, for example. This makes it possible to expand the area of the electrode **20**.

(40) The thickness of the metal film **30** is preferably 1 μm or less. This makes it possible to reduce the possibility of the metal film **30** extending during cleaving. The thickness of the metal film **30** can be 0.1 μm or greater. The metal film **30** has one or more layer of a layer of one or more metal such as Ni, Rh, Cr, Au, W, Pt, Ti, Al or an alloy. The metal film **30** uses an Au layer as the outermost surface, for example. The electrode **20** and the metal film **30** can have the same material. The electrode **20** and the metal film **20** can have the same layering structure. This makes it possible to form the electrode **20** and the metal film **30** in the same step, and possible to simplify the manufacturing process. The metal film **30** is sufficient provided it is constituted using a material for which image recognition is possible. For this reason, instead of the metal film **30**, for example, it is possible to form a dielectric film that reflects the illumination light when performing image recognition in the same manner as the metal film **30**. This kind of dielectric film may have multiple layers, or may have a single layer.

(41) In FIG. **1**, the metal films **30** are provided on the first substrate main surface **16a** of the substrate **16**, but it is also possible to provide the metal films **30** on the semiconductor layer **17**. In this case, the surface on the side on which the ridge **17d** of the semiconductor layer **17** is provided is the first main surface.

(42) (Second Metal Film **40**)

(43) The semiconductor laser element **100** may also have a second metal film **40** provided on the first main surface **11**. The second metal film **40** is provided at a position spaced apart from the electrode **20**, and is in contact with the second side **11b** of the first main surface **11**. As described later, to form the metal film **30** (first metal film) to be reliably in contact with the first side **11a**, it is preferable to obtain the light emitting end surface **13** by dividing the metal film **30** at a certain position. In this case, one of two surfaces obtained by a single division is used as the light emitting end surface **13**, and the other as the light reflecting end surface **14**, with the second metal film **40** formed as shown in FIG. **1**. Alternatively, in a case when using both of the two surfaces obtained by a single division are used as the light emitting end surface **13**, it is possible to not provide the second metal film **40**.

(44) By having the second metal film **40**, it is possible to reduce the possibility of the end surface protective film **62** shown in FIG. **3** described later from reaching the electrode **20**. In this case, it is preferable that the first main surface **11** be a flat plane, and that the thickness of the second metal film **40** and the thickness of the electrode **20** are equal, or that the thickness of the second metal film **40** is greater. This makes it possible to further reduce the possibility of the end surface protective film **62** provided on the light reflecting end surface **14** from reaching the electrode **20**. Particularly, when arranging a spacer on the side of the first main surface **11** when forming the end surface protective film **62**, if the thickness of the second metal film **40** and the thickness of the electrode **20** are in this kind of relationship, it is possible to further reduce wraparound of the end surface protective film **62** using the second metal film **40** and the spacer.

(45) (Second Electrode **50**)

(46) The semiconductor laser element **100** can have a second electrode **50** provided on the second main surface **12**. The second electrode **50** can have a contact electrode **51** and a pad electrode **52**. The second electrode **50** is a p electrode, for example. The second electrode **50** is not in contact with the outer edge of the second main surface **12**, for example. Or, the contact electrode **51** is in contact with the outer edge of the second main surface **12**, but the pad electrode **52** is not in contact with the outer edge of the second main surface **12**.

(47) In FIG. 2A and FIG. 2B, the ridge **17d** is provided on a portion of the semiconductor layer **17**, and the contact electrode **51** is in contact with the ridge **17d**. The contact electrode **51** has one or more layers of an electrically conductive oxide including at least one selected from a metal of Ni, Rh, Cr, Au, W, Pt, Ti, Al, etc. or their alloys, and Zn, In, Sn, for example. As an example of the electrically conductive oxide, an example includes ITO (Indium Tin Oxide), for example. The pad electrode **52** has one or more layers of a layer of a metal of Ni, Rh, Cr, Au, W, Pt, Ti, Al or their alloy, etc., for example. The pad electrode **52** is a layer including Au such as one with the outermost surface being an Au layer, etc., for example.

(48) The electrode **20** (first electrode) and the second electrode **50** may be provided on the same side. For example, it is possible to provide on the surface of the semiconductor layer **17** a portion at which the first electrically conductive semiconductor layer **17a** is exposed and a portion at which the second electrically conductive semiconductor layer **17c** is exposed, to form the electrode **20** on the part at which the first electrically conductive semiconductor layer **17a** is exposed, and to form the second electrode **50** on the part at which the second electrically conductive semiconductor layer **17c** is exposed. In this case, it is possible to form the metal film **30**, the electrode **20**, and the second electrode **50** on the same side of the semiconductor-containing part **10**.

(49) (Other Members)

(50) As shown in FIG. 2A and FIG. 2B, the semiconductor laser element **100** may also have an insulating film **61**. The insulating film **61** is provided on the surface of the semiconductor layer **17**. A portion of the insulating film **61** and a portion of the second electrode **50** are in contact. As shown in FIG. 3, the semiconductor laser element **100** may also have the end surface protective film **62** provided on both or one of the light emitting end surface **13** and the light reflecting end surface **14**. FIG. 3 is a schematic plan view showing another example of the semiconductor laser element **100** of an embodiment. The reflectivity of the end surface protective film **62** provided on the light emitting end surface **13** can be made lower than the reflectivity of the end surface protective film **62** provided on the light reflecting end surface **14**. The mode for using the semiconductor laser element **100** is not limited to the mode of using upper part of each drawing as the top, and can use any direction as the top.

(51) (Light Emitting Device **200**)

(52) FIG. 4A is a schematic plan view showing a light emitting device **200** of an embodiment. The light emitting device **200** has the semiconductor laser element **100**, and a base **70** to which the semiconductor laser element **100** is fixed. The semiconductor laser element **100** is fixed to the base **70** in an arrangement with the second electrode **50** facing the base **70**. The base **70** may also be electrically connected to the semiconductor laser element **100**. The base **70** is a sub mount, for example. The base **70** may also be a package.

(53) As shown in FIG. 4A, by fixing the side of the second main surface **12** to the base **70**, it is possible to confirm the metal film **30** in a state with the semiconductor laser element **100** fixed to the base **70**. This makes it possible to use the metal film **30** in a state with the semiconductor laser element **100** fixed to the base **70** as a reference for pass/fail determination.

(54) FIG. 4B is a schematic cross section view showing another example of the light emitting device **200** of an embodiment. The light emitting device **200** has the semiconductor laser element **100**, the base **70** to which the semiconductor laser element **100** is fixed, a package **81**, and a reflective member **82**.

(55) The base **70** shown in FIG. 4B is a sub mount. The base **70** has an insulating body, and an electrically conductive layer provided on the surface of that body. The insulating body is ceramic, for example. The second electrode **50** of the semiconductor laser element is joined to the electrically conductive layer of the base **70** with an electrically conductive adhesive agent **83** interposed.

(56) The package **81** consists of an electrically conductive material such as a metal, an insulating material such as glass or ceramic, or a composite material of those. The package **81** may be

configured by a combination of a plurality of parts. The package **81** has on one part a light transmissive window that transmits laser light. The laser light emitted by the semiconductor laser element **100** is reflected by the reflective member **82**, and is emitted to outside the light emitting device **200** by passing through the window of the package **81**. The reflective member **82** is unnecessary depending on the position of the window of the package **81**. The light emitting device **200** may also have a wavelength conversion member such as a phosphor, etc., provided at the position through which the laser light passes. The package **81** airtightly seals the semiconductor laser element **100**. The package **81** has an anode electrode and a cathode electrode. The anode electrode of the semiconductor laser element **100** is electrically connected to the anode electrode of the package with an electrically conductive layer or linear wire, etc., interposed. The cathode electrode of the semiconductor laser element **100** is electrically connected to the cathode electrode of the package with an electrically conductive layer or a linear wire, etc., interposed.

(57) As shown in FIG. **4B**, when the side of the semiconductor layer **17** is fixed to the base **70**, compared to when the side of the substrate **16** is fixed to the base **70**, the distance from the optical waveguide **15** to the base **70** is smaller, and the probability that the laser light emitted by the semiconductor laser element **100** will be blocked by the base **70** is high. For this reason, the arrangement shown in FIG. **4B** has a smaller allowable range for the distance between the end part of the base **70** and the end part of the semiconductor laser element **100**. With the light emitting device of the present embodiment, by providing the metal film **30**, it is possible to improve the precision of the pass/fail determination, so it is possible to expect an improvement in yield.

(58) (Method for Manufacturing a Semiconductor Laser Element)

(59) FIG. **5** is a flow chart showing the method for manufacturing the semiconductor laser element **100** of an embodiment. FIG. **6** to FIG. **10** are schematic plan views or cross section views showing the method for manufacturing the semiconductor laser element **100** of an embodiment. The method for manufacturing the semiconductor laser element **100** has a semiconductor-containing part preparation step **S101**, an electrode and metal film forming step **S102**, and a cleaving step **S103**. For the shape, arrangement, material, etc., of each member in the method for manufacturing the semiconductor laser element **100**, the items described above can be used.

(60) (Semiconductor-Containing Part Preparation Step **S101**)

(61) In the semiconductor-containing part preparation step **S101**, as shown in FIG. **6** and FIG. **7**, the semiconductor-containing part **10** having the first main surface **11**, the second main surface **12**, and the optical waveguide **15** is prepared. FIG. **6** is a schematic plan view showing the method for manufacturing the semiconductor laser element of an embodiment. FIG. **7** is a cross section view in line VII-VII of FIG. **6**. The distance between the first main surface **11** and the active layer **17b** is greater than the distance between the second main surface **12** and the active layer **17b**. Because of this, it is possible to say that the distance between the first main surface **11** and the optical waveguide **15** is greater than the distance between the second main surface **12** and the optical waveguide **15**. The semiconductor-containing part **10** can have the substrate **16** and the semiconductor layer **17**. The semiconductor-containing part **10** has a wafer shape, for example. The semiconductor layer **17** can be formed on the surface of the substrate **16** using a metal organic vapor deposition method (MOCVD method), for example. The ridge **17d** can be formed using etching, for example.

(62) (Electrode and Metal Film Forming Step **S102**)

(63) In the electrode and metal film forming step **S102**, as shown in FIG. **8** and FIG. **9**, a plurality of electrodes **20** and metal films **31** are formed on the first main surface **11** of the semiconductor-containing part **10**. FIG. **8** is a schematic plan view showing the method for manufacturing a semiconductor laser element of an embodiment. FIG. **9** is a cross section view in line IX-IX of FIG. **8**. The plurality of electrodes **20** are formed aligned in a first direction Y. The metal films **31** are formed at positions between the plurality of electrodes that are also spaced apart from the plurality of electrodes **20**. Specifically, one or more metal films **31** are formed between adjacent

electrodes **20**. In the metal film **31**, the length in a second direction X intersecting the first direction Y is smaller than the length in the second direction X of the electrode **20**. The metal films **31** are formed at positions that do not overlap with the optical waveguide **15** in the plan view when seen along the normal direction of the first main surface **11**. The electrodes **20** and the metal films **31** can be formed using sputtering or a chemical vapor deposition method (CVD method), for example.

(64) (Cleaving Step **S103**)

(65) In the cleaving step **S103**, as shown in FIG. **10**, cleaving is done between the plurality of electrodes **20**, along the second direction X, at a position that divides the metal film **31**. FIG. **10** is a schematic plan view showing the method for manufacturing a semiconductor laser element of an embodiment. Cleaving can be done, for example, by first forming a groove using a laser scribing device at a portion of the semiconductor-containing part **10**, and after that, by pressing the semiconductor-containing part **10** using a blade. The groove formed using the laser scribing device may also be formed only outside the region that becomes the semiconductor laser element **100**. For example, it is possible to form the groove on one end part of a wafer or a laminate that is a divided piece of the divided wafer, and to cleave the laminate using external force along that groove.

(66) By performing the cleaving step **S103**, the light emitting end surface **13** and the light reflecting end surface **14** are formed. In FIG. **10**, of the two surfaces obtained by cleaving once, one is the light emitting end surface **13**, and the other is the light reflecting end surface **14**. By performing the cleaving step **S103**, the metal film **31** is divided in two, with one being the metal film **30**, and the other being the second metal film **40**. When both of the two surfaces obtained by cleaving once are the light emitting end surface **13**, one metal film **31** becomes two metal films **30**. After the cleaving step, it is also possible to form an end surface protective film **62** on one or both of the light emitting end surface **13** and the light reflecting end surface **14**.

(67) Using the steps above, it is possible to obtain the semiconductor laser element **100** that has the metal films **30**, and that can improve pass/fail determination precision. After the cleaving step **S103**, a dividing step can be performed. In the dividing step, the laminate that passed through the cleaving step **S103** is divided along a planned division line. The planned division line is the intersection line of the light emitting end surface **13** and the light reflecting end surface **14** obtained by cleaving. The division can be performed, for example, by first forming a groove using a laser scribing device at a position overlapping the planned division line, and after that, by pressing the laminate using a blade. The order of the dividing step and the cleaving step **S103** can be changed. A step of forming a member such as a protective film can be performed before the cleaving step **S103**.

(68) (Method for Manufacturing a Light Emitting Device)

(69) FIG. **11** is a flow chart showing the method for manufacturing the light emitting device **200** of an embodiment. As shown in FIG. **11**, the method for manufacturing the light emitting device **200** shown in FIG. **4A** or FIG. **4B** has a step of preparing the semiconductor laser element **100** (semiconductor laser element preparation step **S201**), and a step of fixing the semiconductor laser element **100** to the base **70** (fixing step **S202**). In the fixing step **S202**, the semiconductor laser element **100** is fixed to the base **70** in an arrangement with the second main surface **12** facing the base **70**. The fixing can be performed by interposing an adhesive agent **83**.

(70) After the fixing step **S202**, it is possible to also have a detection step **S203** and a pass/fail determination step **S204**. In the detection step **S203**, detected is the outline of a recognition object that includes the metal film **30** of the semiconductor laser element **100**, the end part of the base **70**, or a part that can be estimated to be the end part. The recognition object can be only these, or may include members other than these. Detection of the outline of the recognition object may also be said to be image recognition of the recognition object. In the pass/fail determination step **S204**, pass/fail determination is performed using the detected outline of the recognition object. The pass/fail determination can be performed by determining whether the distance between the end part

of the base **70** or the part that can be estimated to be the end part and the metal film **30** is within a reference value range, for example. The part that can be estimated as the end part of the base **70** is, for example, a metal layer provided near the end part though it does not match with the end part in the plan view. The distance between the metal layer of the base **70** and the end part of the base **70** is 70 μm or less, for example.

(71) Above, embodiments of the present invention were explained while referring to drawings. However, the present invention is not limited to these specific examples. All modes that can be implemented using suitable design changes by a person skilled in the art based on the embodiments of the present invention described above also are within the scope of the present invention as long as they contain the gist of the present invention. In addition, within the scope of the concepts of the present invention, there are various modified examples and amended examples that can be thought of by a person skilled in the art, and these modified examples and amended examples are understood to belong to the scope of the present invention.

Claims

1. A semiconductor laser element comprising: a semiconductor-containing part having a first main surface, a second main surface, a light emitting end surface, a light reflecting end surface, and an optical waveguide, a distance between the first main surface and the optical waveguide being greater than a distance between the second main surface and the optical waveguide; an electrode provided on the first main surface; and at least one metal film provided on the first main surface at a position spaced apart from the electrode, wherein in a plan view seen along a normal direction of the main surface, an outer edge of the first main surface has a first side on a side of the light emitting end surface, the at least one metal film is in contact with the first side of the outer edge of the first main surface, and the at least one metal film is arranged at a position that does not overlap the optical waveguide.
2. The semiconductor laser element according to claim 1, wherein at least a portion of the electrode is arranged at a position that overlaps the optical waveguide in the plan view.
3. The semiconductor laser element according to claim 1, wherein a length of the at least one metal film along the first side is smaller than a length of the electrode in a direction parallel to the first side.
4. The semiconductor laser element according to claim 1, wherein a length of the at least one metal film along the first side is 70% or less than a length of the electrode in a direction parallel to the first side.
5. The semiconductor laser element according to claim 1, further comprising a plurality of metal films including the at least one metal film, each of the metal films being provided on the first main surface at a position spaced apart from the electrode.
6. The semiconductor laser element according to claim 5, wherein a total length of the metal films along the first side is 30% or greater than a length of the first side.
7. The semiconductor laser element according to claim 5, wherein two of the metal films are respectively arranged at positions sandwiching the optical waveguide in the plan view.
8. The semiconductor laser element according to claim 5, further comprising at least one second metal film provided on the first main surface at a position spaced apart from the electrode, and in contact with a second side of the outer edge of the first main surface on a side of the light reflecting end surface.
9. The semiconductor laser element according to claim 1, wherein the semiconductor-containing part has a substrate that has a first substrate main surface constituting the first main surface, and a second substrate main surface on a side opposite to the first substrate main surface, and a semiconductor layer provided on the second substrate main surface of the substrate.
10. The semiconductor laser element according to claim 9, further comprising a second electrode

provided on the second main surface.

11. The semiconductor laser element according to claim 1, further comprising a protective film covering at least one of the light emitting end surface and the light reflecting end surface of the semiconductor-containing part.

12. A light emitting device comprising: the semiconductor laser element according to claim 1; and a base on which the semiconductor laser element is fixed with the second main surface of the semiconductor-containing part facing the base.

13. A semiconductor laser element comprising: a semiconductor-containing part having a first main surface, a second main surface, a light emitting end surface, a light reflecting end surface, and an optical waveguide, a distance between the first main surface and the optical waveguide being greater than a distance between the second main surface and the optical waveguide; an electrode provided on the first main surface at a position spaced apart from the light emitting end surface in a plan view seen along a normal direction of the first main surface; and at least one metal film provided on the first main surface at a position spaced apart from the electrode and the optical waveguide in the plan view, an edge of the at least one metal film being flush with the light emitting end surface in the plan view.

14. The semiconductor laser element according to claim 13, further comprising a plurality of metal films including the at least one metal film, each of the metal films being provided on the first main surface at a position spaced apart from the electrode and the optical waveguide in the plan view, and an edge of each of the metal films being flush with the light emitting end surface in the plan view.

15. The semiconductor laser element according to claim 14, further comprising at least one second metal film provided on the first main surface at a position spaced apart from the electrode, and an edge of the at least one second metal film being flush with the light reflecting end surface in the plan view.

16. A light emitting device comprising: the semiconductor laser element according to claim 13; and a base on which the semiconductor laser element is fixed with the second main surface of the semiconductor-containing part facing the base.
